

Letter of Submittal

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Steven Winiski
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3831 No. 6 Road
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Dear Mr. Winiski,

Please accept this technical report summarizing the ignition SCADA transition project I worked on during my Fall 2025 Co-op term at Dometic Vancouver. The project focuses on digitizing machine and workbench information and building a structured preventive maintenance (PM) system that will ultimately be managed through SCADA.

This report describes the motivation for the project, the data acquisition and verification methods used across all three buildings, the challenges posed by legacy paper and spreadsheet-based records, and the planned structure of the future SCADA-driven maintenance workflow. The work was conducted at the Dometic Vancouver facility in Richmond, BC, located on the traditional and unceded territories of the Musqueam, Squamish, and Tsleil-Waututh Nations.

The report is original and has not been submitted for credit elsewhere. I worked under your supervision as my manager, and collaborated closely with Process Development Engineer Huseyin Doganay, the SCADA project lead, and Operations Engineering Manager Rob Pehl, who provided guidance on machine and bench naming, work cell structures, and PM validation.

Sincerely,

Ryan Thomas

Ignition SCADA Transition Project

Technical Report - Fall 2025 Co-op Term

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Abstract

This report documents the development of a digital maintenance data foundation to support a future SCADA-based preventive maintenance system at Dometic Vancouver. The facility consists of three buildings containing a wide range of machines and workbenches, including CNCs, Drakes, lathes, and various custom workstations. Historically, machine maintenance, cleaning, and repair activities have been tracked on paper logs and in scattered spreadsheets, which limits real-time visibility, hampers data collection, and makes automated scheduling impossible.

The SCADA project aims to digitize machine and workbench information and fully automate the process of tracking and assigning PMs, cleanings, and repairs. My role focused on acquiring, understanding, verifying, and organizing large amounts of asset and maintenance-related data including machines, work centers, PMs, PM frequencies, cleaning tasks, oil changes, maintenance personnel, groups, contractors, and more, and transitioning the information into a structure suitable for SCADA integration.

The methodology included defining a data model, consolidating existing digital records, planning and executing a facility-wide sweep across all three buildings to verify asset identities and locations, organizing assets into work cells, and linking each asset to appropriate PM tasks. A key challenge was the amount of outdated and inconsistent information in legacy documents, requiring in-person verification and renaming under the guidance of Operations Manager Rob Pehl.

The report also outlines the envisioned SCADA workflow: automatic weekly generation of maintenance schedules, email notifications to the correct personnel based on asset and group assignments, clear instructions attached to specific PM tasks, and digital feedback loops when maintenance is completed. The outcomes of this project establish a robust foundation for Dometic Vancouver's transition from paper-based maintenance tracking to an automated, SCADA-driven system.

Introduction

Preventive maintenance is essential in a manufacturing environment where machines and workbenches directly support production output and safety. At Dometic Vancouver, PM activities such as cleaning, lubrication, repairs, testing, and inspections have historically been recorded on paper sheets and maintained in multiple, partially overlapping spreadsheets. This approach works at a basic level but has several limitations:

- No real-time visibility into when a machine was last cleaned or serviced
- Difficulty identifying missed cycles or overdue tasks
- Time-consuming manual sorting and assignment of maintenance tasks
- High risk of errors due to outdated or inconsistent information

The SCADA maintenance project is intended to address these limitations by digitizing machine and workbench information and creating a system that automatically schedules and assigns PMs, sends notifications, and records completion in real time. For this to be possible, the underlying data for machines, benches, work cells, PM tasks, frequencies, and responsible personnel must be accurate, complete, and structured.

This report focuses on building that data foundation, describing the process from raw legacy information to a SCADA-ready maintenance dataset.

Problem Identification & Objectives

Limitations of Paper-Based and Fragmented Digital Records

The starting point for the project was a mixture of:

- Paper logs for machine cleaning and maintenance
- Inaccessibility of reliable information in a timely fashion
- Spreadsheets with partial or outdated lists of PM's for machines and benches
- A multitude of benches with outdated or incorrect names and numbers

In this system, if someone wants to know when a machine was last cleaned or what specific work was done, they must manually backtrack through paper copies and spreadsheets. There is no ability to automatically calculate when the next PM is due or to generate a clear overview of maintenance load for the coming week or the reliability of the PMs if questions are raised.

Furthermore, because the facility has evolved over many years, machines moved, bench layouts changed, and some equipment was decommissioned, many records were outdated in the midst of the fast paced environment. This included incorrectly named machines, missing or decommissioned benches still listed as active, and PMs linked to equipment that no longer existed.

Project Objectives

The SCADA maintenance data foundation project had several specific objectives:

1. Digitize and Consolidate Information:
Collect all existing machine and workbench information from paper and spreadsheets into one structured dataset.
2. Verify and Correct Asset Information:
Perform a physical sweep of all three buildings to verify that each bench and machine is correctly named, numbered, and located, and classify assets as active, spare, or decommissioned. Verify PMs with machinists and bench designers.
3. Organize Assets into Work Cells:
Assign each bench and machine to a parent work cell and building, creating a logical hierarchy that SCADA can use to group and filter assets.
4. Link PM Tasks to Assets:
Match PM tasks (e.g., cleanings, oil changes, inspections) and their frequencies to the

correct machines and workbenches.

5. Design a SCADA-Based Workflow:

Structure the data so that SCADA can automatically generate weekly maintenance schedules, send emails to the right people or groups, and store feedback when maintenance is completed.

My responsibilities were primarily centered on data acquisition, verification, organization, and preparing this information for SCADA upload.

Methodology

Data Model and Scope of Information

Before starting the detailed work, we defined what information SCADA would eventually need to operate effectively. For each asset (Machine or Bench), the following data elements were considered essential:

- Unique asset ID (parent classes, child classes, Work or Machine Center, Cells, Groups)
- Descriptive asset name and number (eg. MC_522 Drake, WC_2010-01 SPA EOL Test)
- Asset type (e.g., CNC, lathe, bench, station)
- Parent work cell and building
- Operational status (active, spare, decommissioned)
- List of associated PM tasks with frequencies (e.g., every 7 days, monthly, quarterly)
- Linked personnel, groups, or contractors responsible for the maintenance

In order to create this the information required beyond the assets themselves are:

- PM types/tasks (cleaning, oil change, inspection, adjustment, removing chips, etc.)
- PM frequencies and triggers
- Maintenance personnel and groups
- Outsourced maintenance contractors

Although this would not be considered “big data” in the usual sense, the scope is large for a single facility: dozens of machines, numerous work cells, multiple buildings, and a long list of PM tasks and personnel relationships.

Data Acquisition and Initial Organization

The first phase was to gather everything that already existed. I collected:

- Pre existing “legacy” Machine and Work Bench Lists

- Paper-based machine cleaning and maintenance logs
- Existing PM spreadsheets, schedules, and templates
- Existing PM instructions and tutorials

With this plan outlined, we then used Ignition, a software platform designed for SCADA applications such as process control and data management, to create the initial logic for how the program would work. We called it the Asset Management System, and we organized information under the following categories:

- Asset Register: Contains all machines and workbenches, including names, numbers, parent IDs, hierarchies, manufacturers, model specifications, serial numbers, years, building assignments, status, and notes.
- Maintenance types: Stores all collected PMs, assigns each PM a unique ID, attaches instruction links when necessary, and displays PM status.
- Personnel: Includes all in-house workers such as maintenance personnel and machinists.
- Personnel groups: Defines groups required when multiple people must complete a task (e.g., the maintenance group).
- Outsource companies: Lists external contractors and identifies PM tasks that cannot be performed in-house.
- Maintenance Schedule: The main screen where all information integrates. For each scheduled PM, it displays: asset ID, asset name, maintenance type, maintenance description, assigned personnel, assigned group, outsourced company (if applicable), frequency, next due date, and completion status.

Physical Verification: Facility Sweep Across Three Buildings

The main challenge in the project was verifying the integrity of the information in Dometic's files. Because the facility has been in operation for a long time and has undergone many physical changes and movements, we found a significant amount of outdated or incomplete data when coming into contact with PM's and especially work benches that weren't accounted for during moves and therefore did not have or had incorrect names, numbers, and descriptions.

To address this, we planned and executed a full sweep of all three warehouses to account for every machine and work bench to ensure accurate and up to date data. Working closely with Operations Manager Rob Pehl, who knows the bench and machine names and locations by heart, we visited every bench, machine, and station in each building.

For each asset, we:

- Confirmed the actual name or identifier number used on the floor
- Checked whether it matched the name and number in the dataset
- Verified whether the asset was active, spare, or decommissioned
- Adjusted naming and numbering as needed, following conventions approved by Rob
- Ensured that each bench belonged to a clearly defined parent work cell

The outcome of this sweep is not just a list of benches and machines with correct naming; it is a significantly more accurate, reliable, and updated view of the entire facility layout.

Organizing Assets into Work Cells and Hierarchies

Once bench and machine information was verified, I organized them into a hierarchical structure. At a high level, the hierarchy can be described as:

- Level 1: Work Center (WC) or Machining Center (MC)
- Level 2: Parent Cell (System) or Child Cell (Component)
- Level 3: Individual asset (machine or bench) named and numbered eg. WC (level 1)_2010-01 (Component, 2010 is the System) SPA EOL Test (Individual Asset)

Within each work cell, benches and machines were grouped logically according to their function and their physical location. This structure is critical for SCADA, since it allows maintenance to be viewed and assigned by area, department, or specific machine.

Figure 1. Placeholder – Example of asset hierarchy by building, work cell, and asset.

Linking PM Tasks, Frequencies, and Instructions

After the assets were structured, the next step was to ensure that every active machine and bench had appropriate PM tasks assigned. We focused on PM tasks that occur once every

seven days or longer (weekly, monthly, quarterly, etc.), as these are especially suitable for automated scheduling.

To do this, we will collaborate with multiple engineering groups around the facility, including:

- Machining engineers
- Engineers responsible for bench creation and modification
- Manufacturing and quality engineering departments

The process includes:

1. Reviewing existing PM lists and matching them to the correct assets.
2. Revising PM frequencies where they no longer made sense given current usage.
3. Identifying PM tasks that require additional written instructions for clarity and consistency.
4. Drafting or refining those instructions in consultation with the relevant engineers.

Once this is complete, each asset will have a verified list of PM tasks, each with a frequency and, where needed, instructions explaining exactly what needs to be done and where on the machine or bench it must be performed.

Figure 2. Placeholder – Sample PM assignment structure for one work cell.

Designing the SCADA Maintenance Workflow

With accurate benches, machines, PM tasks, and personnel lists, the SCADA system can be configured to run as an actual automated maintenance platform. The planned workflow is:

1. Weekly Schedule Generation:
Every Thursday, the system will automatically generate the required PM tasks for the following week, based on asset-specific frequencies and rules.
2. Email Notifications:
The system will send emails or notifications to the correct people, groups, or contractors responsible for the maintenance of related machines or workbenches.
3. Clear Identification and Instructions:
Because the asset naming conventions and work cell structures will be consistent, maintenance personnel will be able to quickly identify the correct location of each

machine or bench. If a PM task has associated instructions, these will be included in the notification so workers know exactly what needs to be done.

4. Completion Feedback:

After maintenance is performed, the responsible person will interact with the system (e.g., confirming completion in the SCADA interface). This feedback will need to be approved then stored in the system's memory, building a digital history of maintenance for each asset.

5. Long-Term Data Access:

At any point, supervisors and engineers will be able to access maintenance histories, check when tasks were completed, and review which tasks were overdue or missed.

Figure 3. Placeholder – Conceptual workflow for SCADA-based PM generation and feedback.

Results & Discussion

The project has already produced significant value even before full SCADA deployment. The main results include:

- A consolidated, cleaned, and structured dataset of machines and workbenches across all three buildings.
- Verified naming and numbering, reducing confusion and improving communication between maintenance and operations.
- A clear asset hierarchy that SCADA can use to organize and display maintenance data.
- A defined process for validating PM tasks and frequencies with various engineering groups.
- A conceptual design for the automated weekly maintenance schedule, email notifications, and completion feedback loop.

From a learning perspective, the project also illustrated how even a “small-scale” data environment can behave like a big-data problem when inaccuracies, legacy records, and multiple stakeholders are involved. Success depended not only on Excel and data skills but also on structured planning, attention to detail, and close collaboration with people who know the facility well.

Conclusion & Recommendations

The SCADA Maintenance Data Foundation project is a critical step in moving Dometic Vancouver from paper-based, reactive maintenance tracking to a proactive, automated, and fully digital maintenance system. By systematically acquiring, verifying, and organizing asset and PM data, the project lays the groundwork for SCADA to function effectively.

Key conclusions:

- Accurate data is essential for any automation effort; SCADA can only be as reliable as the asset information it receives.
- Physical verification through a facility sweep is necessary when legacy data is inconsistent or outdated.
- A clear hierarchy of buildings, work cells, and assets makes maintenance easier to visualize and manage.
- Involving multiple engineering and operations stakeholders ensures that PM tasks and instructions are realistic and complete.

Recommendations:

- Continue the PM validation phase with engineering teams to finalize all tasks and instructions.
- Maintain the asset dataset as a living document, updating it whenever equipment is moved, added, or removed.
- Implement the SCADA maintenance workflow and monitor its performance, adjusting frequencies and instructions based on real feedback.
- Train maintenance personnel and future Co-op students on the naming conventions and data structure so the system remains consistent over time.

This project opened the door for me to experience what handling big data was like and significantly developed my skills in structured data handling, system thinking, and

cross-functional communication, and it provides a strong foundation for the next phases of SCADA implementation at Dometic Vancouver.